

PG($k - 1, 2$) Geometry as a Mathematical Framework for Order-Dependent Processes in Biology: Drug Metabolism and the Genetic Code

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Abstract. We propose that finite projective geometries PG($k - 1, 2$) provide a natural mathematical framework for analysing order-dependent processes in biology. Two applications are developed. First, we observe that the FDA’s canonical set of seven major drug-metabolising CYP450 isoforms admits a bijection with the points of the Fano plane PG(2, 2), under which the 343 ordered enzyme triples partition into 133 trivial, 42 associative (corresponding to genetically or functionally clustered enzyme families), and 168 non-associative triples. We derive testable predictions: drug–drug interactions mediated by enzyme triples on a Fano line should exhibit weaker order-dependence than those off a Fano line. Second, we embed the 64 codons of the standard genetic code into $\mathbb{Z}_2^6$ via the biochemical binary encoding (purine/pyrimidine $\times$ weak/strong hydrogen bond) and compute properties of the resulting PG(5, 2) structure. Hamming distance in $\mathbb{Z}_2^6$ correlates monotonically with amino acid hydrophobicity difference ($r = 0.199$, strictly monotonic across all six distance classes), and 56% of single-bit mutations preserve amino acid chemical class. The ambient Cayley–Dickson algebra of dimension 64 has $T_6 = 130\,200 = 775 \times 168$ nonzero basis associator triples [1], confirming that 168 is the quantum of non-associativity at the dimension where the genetic code lives. Validation against clinical pharmacovigilance data is beyond the scope of this mathematical note and is left as empirical future work.

Keywords: CYP450, drug–drug interactions, genetic code, projective geometry, Fano plane, non-associativity, Cayley–Dickson

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1. Introduction

Many biological processes are order-dependent: the clinical outcome of a drug cascade depends on the sequence of administration [2], the phenotypic effect of a mutation path depends on the order of substitutions [3], and the efficacy of combination therapies depends on scheduling [4]. Despite this, mathematical frameworks for classifying *which* combinations are order-dependent and which are not remain underdeveloped.

In algebra, order-dependence is captured by the **associator** $[a, b, c] = (ab)c - a(bc)$. When the associator vanishes, the parenthesisation of a triple product does not matter; when it is nonzero, the order of operations is significant. The octonions $\mathbb{O}$ — the largest normed division algebra — are the first algebra in the Cayley–Dickson tower where the associator is generically nonzero. The count of nonzero basis associator triples is exactly $168 = |\text{PSL}(2, 7)|$, governed by the Fano plane PG(2, 2) [5, 6].

In a companion paper [1], we showed that this 168-divisibility extends throughout the Cayley–Dickson tower, with $T_k = 168(P_k - 4P_{k-1})$ verified at four levels through dimension 64.

In this paper, we ask: *do biological systems whose interaction structure involves 7 or $2^k - 1$ elements exhibit non-associative patterns consistent with the $\text{PG}(k - 1, 2)$ geometry?* We develop two applications — one at the scale of $\text{PG}(2, 2)$ (drug metabolism) and one at $\text{PG}(5, 2)$ (the genetic code) — and derive testable predictions from each.

2. The CYP450 system in $\text{PG}(2, 2)$

2.1. Background: CYP450 drug metabolism

The cytochrome P450 (CYP) superfamily is responsible for the oxidative metabolism of the majority of clinically used drugs [7]. Regulatory agencies identify seven principal isoforms for drug–drug interaction (DDI) assessment: CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, and CYP3A4 [8].

Drug interactions mediated by these enzymes are frequently order-dependent. Mechanism-based inhibition (MBI) — irreversible inactivation of the enzyme — creates a time-arrow: pre-treatment with an MBI perpetrator before introduction of a victim substrate produces maximum inhibition from day 1, whereas the reverse order produces a gradual escalation over 3–5 days [9]. Quantitatively, 67% of pharmacodynamic DDIs in a large-scale analysis were found to be monodirectional, with distinct perpetrators and victims [2].

2.2. The bijection

Under the FDA’s canonical seven-isoform simplification, the CYP450 system admits a bijection with the points of $\text{PG}(2, 2)$. We propose the following mapping, motivated by genetic clustering and substrate overlap:

Basis	CYP isoform	Chromosome	Rationale
e_1	CYP1A2	15q24	Isolated family; planar aromatic substrates
e_2	CYP2C9	10q23	CYP2C cluster; acidic/neutral substrates
e_3	CYP2C8	10q23	CYP2C cluster; shared substrates with CYP3A4
e_4	CYP2B6	19q13	CYP2 family; distinct from CYP2C
e_5	CYP2C19	10q23	CYP2C cluster; completes Fano line (2, 3, 5)
e_6	CYP2D6	22q13	Unique: non-inducible; basic nitrogen substrates
e_7	CYP3A4	7q22	Dominant enzyme ($\approx 30\%$ of drugs)

Table 1: Proposed bijection between CYP450 isoforms and $\text{PG}(2, 2)$ points.

The mapping is constrained by the requirement that the three enzymes of the CYP2C gene cluster on chromosome 10q23.33 — CYP2C8, CYP2C9, and CYP2C19 — form a Fano line (here, line (2, 3, 5)). These three enzymes share a genetic locus spanning ≈ 500 kb, exhibit overlapping substrate selectivity, and show coordinated expression patterns [10]. Within the Fano plane, additional lines acquire pharmacological interpretations:

Fano line	CYP isoforms	Pharmacological interpretation
(2, 3, 5)	CYP2C9–CYP2C8–CYP2C19	CYP2C gene cluster (chr 10)
(6, 7, 2)	CYP2D6–CYP3A4–CYP2C9	The “big three” metabolisers ($\approx 65\%$ of drugs)
(4, 5, 7)	CYP2B6–CYP2C19–CYP3A4	Prodrug activation triad
(5, 6, 1)	CYP2C19–CYP2D6–CYP1A2	Polymorphic triad
(7, 1, 3)	CYP3A4–CYP1A2–CYP2C8	Paclitaxel metabolism bridge
(1, 2, 4)	CYP1A2–CYP2C9–CYP2B6	CYP1/2 family axis
(3, 4, 6)	CYP2C8–CYP2B6–CYP2D6	CYP2 subfamily convergence

Table 2: Fano lines and their pharmacological interpretations under the proposed mapping.

2.3. The 168 partition and testable predictions

The 343 ordered triples of CYP isoforms partition as $343 = 133 + 42 + 168$, where 133 involve repeated enzymes (trivially order-independent), 42 lie on Fano lines (predicted order-independent), and 168 are non-collinear (predicted order-dependent). This partition follows algebraically from the Fano plane structure and is independent of the biological mapping.

The biological content lies in the mapping itself: we predict that the 42 Fano-line triples correspond to enzyme combinations whose DDI outcomes are less sensitive to administration order (because they share a mechanistic family), while the 168 non-collinear triples correspond to combinations where order-dependence is clinically significant.

Prediction 1. *DDIs mediated by CYP enzyme triples on a Fano line (e.g., CYP2C9–CYP2C8–CYP2C19) should exhibit weaker order-dependence than DDIs mediated by non-collinear triples (e.g., CYP1A2–CYP2D6–CYP3A4), as measured by asymmetry of AUC ratios in perpetrator–victim pharmacokinetic studies.*

Prediction 2. *Among the 168 non-collinear triples, those involving CYP2D6 (the only non-inducible major isoform) should show the strongest order-dependence, because the lack of induction-mediated recovery creates a longer-lasting time-arrow.*

Prediction 3. *The binary property of the associator norm ($\| [e_i, e_j, e_k] \| \in \{0, 2\}$) predicts a bimodal distribution: DDI order-dependence should be either absent or fully present, with few intermediate cases.*

Validation of these predictions requires analysis of pharmacovigilance databases (e.g., FDA FAERS) or dedicated pharmacokinetic studies with controlled sequencing, and is beyond the scope of this mathematical note.

3. The genetic code in $\text{PG}(5, 2)$

3.1. The biochemical binary encoding

Each nucleotide base has two binary biochemical properties:

Base	Ring	H-bond strength	$\mathbb{Z}_2^2$ vector
A	Purine (0)	Weak (0)	(0, 0)
G	Purine (0)	Strong (1)	(0, 1)
U	Pyrimidine (1)	Weak (0)	(1, 0)
C	Pyrimidine (1)	Strong (1)	(1, 1)

Table 3: Binary encoding of nucleotide bases.

Each codon (three bases) maps to a vector in $\mathbb{Z}_2^6 = \mathbb{Z}_2^2 \times \mathbb{Z}_2^2 \times \mathbb{Z}_2^2$. The 63 nontrivial codons (excluding AAA, the zero vector) are the points of $\text{PG}(5, 2)$. This encoding is not arbitrary — it is determined by the biochemistry of the bases.

3.2. Hamming distance and amino acid chemistry

The relationship between Hamming distance in $\mathbb{Z}_2^6$ and amino acid chemical similarity was studied by Freeland and Hurst [11] from an evolutionary optimality perspective. The present work recasts this relationship in the projective geometry $\text{PG}(5, 2)$ and connects it to the Cayley–Dickson tower via $T_6 = 168 \times 775$ [1].

We compute the mean absolute difference in Kyte–Doolittle hydrophobicity [12] between all pairs of nonzero codons, stratified by Hamming distance:

Hamming distance	Mean $ \Delta H $	N pairs
1	2.03	186
2	3.02	465
3	3.53	620
4	3.78	465
5	3.92	186
6	4.00	31

Table 4: Mean hydrophobicity difference by Hamming distance in $\mathbb{Z}_2^6$. The relationship is strictly monotonic.

The relationship is strictly monotonic: codons that are near in $\mathbb{Z}_2^6$ encode amino acids with similar hydrophobicity. The Pearson correlation between Hamming distance and $|\Delta H|$ is $r = 0.199$ (computed over all 1,953 nonzero codon pairs).

3.3. Mutation robustness

Among 372 single-bit mutations (Hamming distance 1) from nonzero codons, 26.3% are synonymous (same amino acid) and 55.9% preserve amino acid chemical class (nonpolar, polar, positive, negative). The genetic code is robust to the perturbation structure defined by $\mathbb{Z}_2^6$.

3.4. Fano line structure

In contrast to the metric (Hamming distance) structure, Fano lines in $\text{PG}(5, 2)$ do not organise codons by amino acid identity or class. Among 651 Fano lines connecting nonzero codons, 64 (9.8%) have all three codons in the same chemical class, compared to a random baseline of 11.5%. The genetic code’s degeneracy follows the *metric* structure of $\text{PG}(5, 2)$, not its seven-point subplane structure.

3.5. The ambient algebra

The 63 nonzero codons are the imaginary basis elements of the $2^6 = 64$ -dimensional Cayley–Dickson algebra. In [1], we verified exhaustively that $T_6 = 130\,200 = 775 \times 168$: among the $63^3 = 250\,047$ ordered basis triples, exactly 130,200 have nonzero associator, and all nonzero associator norms equal exactly 2. The genetic code lives in a space with 775 copies of the 168-quantum of non-associativity.

4. The fractal branching structure

The Fano subplane measure in the Cayley–Dickson tower proliferates with branching factor 7 per doubling [1], yielding a combinatorial scaling exponent

$$D_{\text{Fano}} = \log_2 7 \approx 2.807 \quad (1)$$

in three-dimensional triple space. The codimension $3 - D_{\text{Fano}} \approx 0.193$ characterises the gap between the Fano subplane skeleton and the ambient triple space.

Biologically, this means that the non-associative structure at each scale of the Cayley–Dickson tower is organised by the same self-similar Fano architecture: $\text{PG}(2, 2)$ at the enzyme scale, $\text{PG}(5, 2)$ at the codon scale, with the 168-quantum preserved at every level and the biology adapting to the relevant geometric structure at each scale — combinatorial (Fano lines) for small systems, metric (Hamming distance) for large ones.

5. Discussion

The framework presented here is a mathematical model, not a claim that biological systems are governed by non-associative algebra. The value of the model lies in its testable predictions and its capacity to organise known phenomena (DDI order-dependence, mutation robustness) within a unified geometric structure.

Several limitations should be noted. The CYP450 mapping requires the FDA’s seven-isoform simplification; in reality, additional CYP enzymes (CYP2E1, CYP3A5, and others) contribute to drug metabolism, and the restriction to seven is a regulatory convention. The genetic code embedding depends on the specific binary encoding (Table 3); other biochemically motivated encodings are possible and may yield different geometric properties. The ambient Cayley–Dickson algebra at dimension 64 is not directly involved in codon biology — it provides the algebraic context in which the projective geometry acquires its associator structure.

The strongest immediate application is the CYP450 model, where the seven-element constraint is externally validated (by regulatory practice), the order-dependence is clinically documented (67% monodirectional DDIs), and the predictions are testable against pharmacovigilance data. We encourage validation using the FDA Adverse Event Reporting System (FAERS) or controlled pharmacokinetic studies with varied drug sequencing.

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